NORTHWIND TRADERS

BUSINESS PERFORMANCE ANALYSIS

Capstone Project Report

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1. Executive Summary

This project analysed Northwind Traders business performance across orders, products, customers, shipping providers and employees. Seven related tables were assessed and modelled in PostgreSQL; SQL was used to investigate business questions, and Tableau Public was used to build an interactive dashboard.

The dataset contained 700 orders and generated estimated revenue of $1 131 114.86, with an average order value of $1 615.88. March 2015 recorded the highest monthly revenue in the analysis, while May 2015 had substantially lower activity. Côte de Blaye was the highest-revenue product. Shipping-provider results differed: Federal Shipping had the shortest average order-to-ship time, while United Package had the longest. Overall, 95.4% of orders were shipped on or before their required dates.

The findings provide management with a clearer view of sales performance, product and customer contributions, and shipping operations. The interactive dashboard (Figure 1) brings these measures together, allowing users to explore performance across different business areas. While the analysis highlights patterns and differences that may warrant further investigation, additional information would be needed to establish their underlying causes, assess product profitability, and evaluate actual delivery performance.

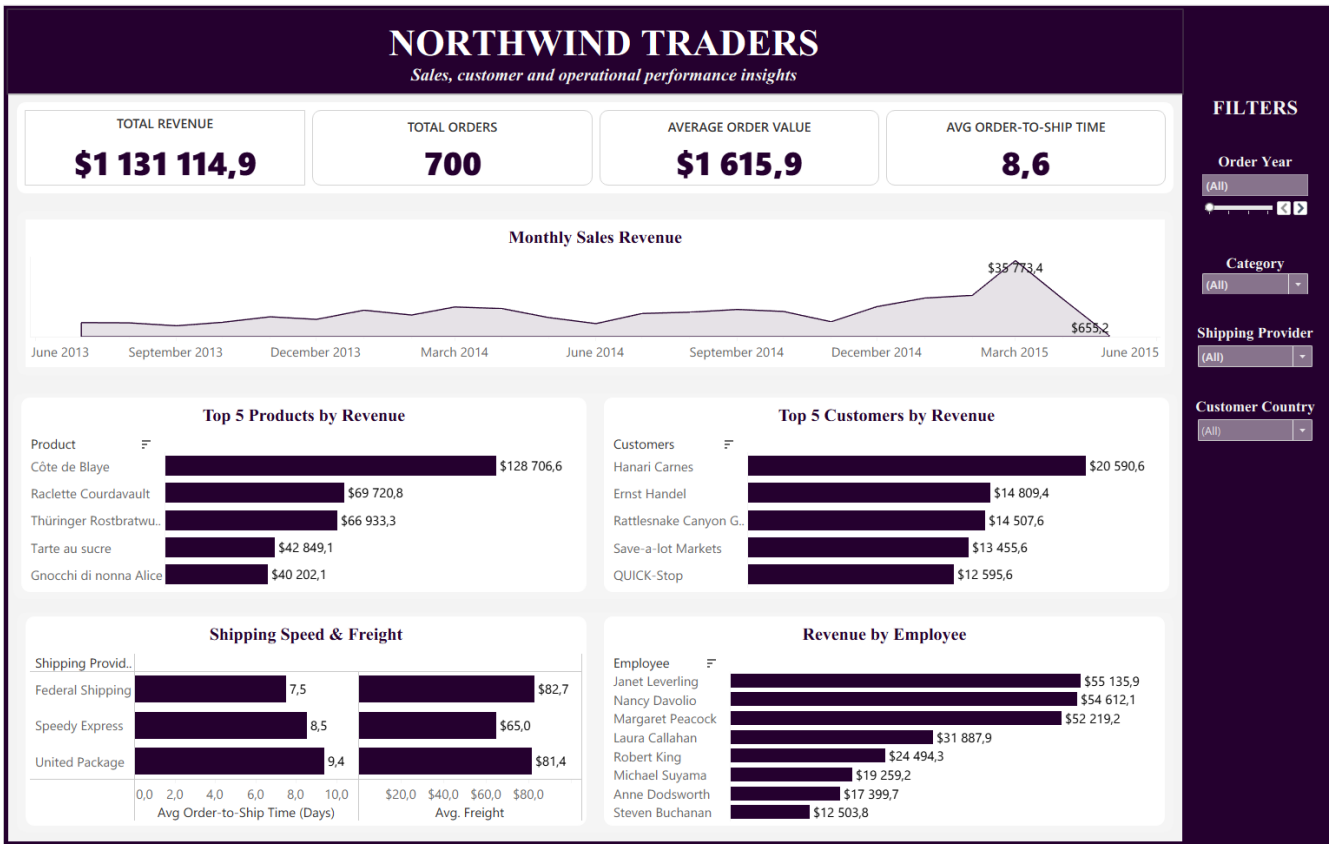


Figure 1: Interactive Northwind Traders Business Performance Dashboard in Tableau Public.

2. Business Context and Project Objectives

2.1 Business context and problem

Northwind Traders is a global supplier of specialty food products. Its operations depend on customer orders, product demand, employees and shipping providers. The project brief identified a need for actionable insight into these areas and required a database, SQL-based analysis and an interactive dashboard.

The business problem addressed in this project was the need for a consolidated view of commercial and operational performance. Management needed to review order and revenue trends, identify products and customers that contribute to sales, compare shipping costs and shipping speed, and examine employee contributions.

2.2 Objectives and business questions

- Analyse order volume and revenue over time.
- Identify the best- and worst-performing products and categories by sales revenue.
- Identify key customers based on order activity and revenue contribution.
- Compare shipping providers on freight charges and order-to-ship time.
- Assess how many orders were shipped on or before their required dates.
- Compare employee contributions across order volume, revenue and fulfilment measures.
- Build an interactive dashboard with the required headline metrics and filters.

3. Dataset and Methodology

3.1 Dataset overview

The supplied dataset contained seven related tables. The record counts below were checked as part of the project.

Table	Records	Purpose
Customers	91	Customer details and geographic information
Orders	700	Order dates, customer, employee, shipper and freight information
Order Details	1 801	Products, quantities and discounts for each order line
Products	77	Product names, unit prices and category assignments
Categories	8	Product category descriptions
Employees	9	Employee details and reporting relationships
Shippers	3	Shipping-provider details

3.2 Data quality and preparation

Before conducting the analysis, the seven tables were reviewed for missing values, duplicate identifiers, invalid values, date inconsistencies and relationship integrity.

The following checks were performed:

- **Missing values:** Checked relevant fields for blanks, noting that the blank ReportsTo value for the top-level employee was structurally valid.
- **Duplicate records:** Checked primary keys and the composite key formed by OrderID and ProductID in the Order Details table.
- **Value validation:** Checked that product unit prices were positive, order-detail quantities were greater than zero and discount values fell within the observed range of 0%–25%.
- **Date validation:** Checked for orders where the required date or shipped date occurred before the order date.
- **Relationship integrity:** Verified that referenced customer, employee, shipper, product and category identifiers existed in their corresponding tables.
- **Data consistency:** Reviewed descriptive packaging information in QuantityPerUnit, which used inconsistent formats and could require standardisation if used in further analysis.

No major data-quality issues were identified during these checks. The dataset was considered suitable for the planned analysis, subject to the limitations documented later in this report.

3.3 Database model and tools

PostgreSQL was used to create the northwind_traders database and define primary and foreign key relationships. The model linked Customers, Employees and Shippers to Orders; Orders to Order Details; Products to Order Details; and Categories to Products. These one-to-many relationships supported analysis at order, order-line, product, customer and shipper levels.

During PostgreSQL import, comma decimal separators in some fields caused numeric conversion errors. The affected fields were imported as text and converted to numeric types using SQL, replacing commas with decimal points where required. Imported row counts were checked against the source workbook.

SQL was used for aggregation and business-question analysis. Tableau Public was used to develop the interactive dashboard. Because the available Tableau Public setup did not provide a direct PostgreSQL connection, the dashboard used the original Excel workbook as its data source, with the seven related tables connected through Tableau relationships. The SQL analysis and dashboard therefore used separate data connections rather than one live integrated source.

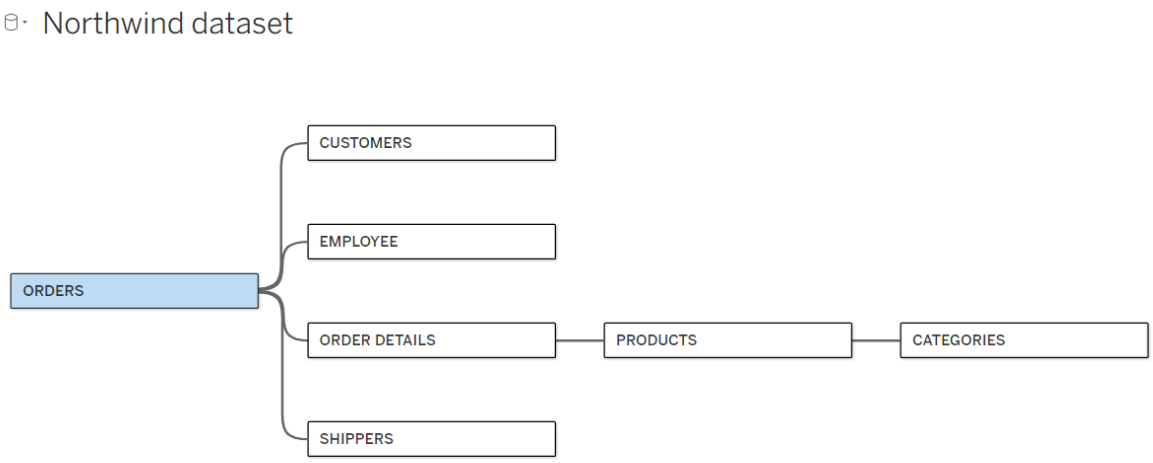


Figure 2: Data Model of the Northwind Traders Dataset

3.4 KPIs and definitions

KPI	Definition used
Total Orders	Count of distinct order IDs.
Total Revenue	Sum of quantity × unit price × (1 − discount), aggregated from order lines.
Average Order Value	Total estimated revenue divided by distinct orders.
Average Order-to-Ship Time	Average number of days between order date and shipped date.

4. Key Findings and Business Implications

4.1 Overall sales performance varied considerably by month

The analysis recorded 700 orders, estimated revenue of $1,131,114.86 and an average order value of $1,615.88. March 2015 recorded the highest monthly revenue, with 64 orders. May 2015 contained only four orders across two dates, 1 and 4 May. As 4 May was the latest order date in the dataset, May represents a partial reporting period. Its lower recorded activity therefore cannot be directly compared with complete months to assess changes in business performance.

Business implication: Management should interpret monthly trends in the context of the available reporting period. A more reliable comparison would require a complete reporting period or confirmation of the dataset's intended coverage.

4.2 Product and customer contributions were uneven

Côte de Blaye was the highest-revenue product. Customer analysis also showed differences in order activity and revenue contribution across the customer base. These findings help identify products and customers that account for a meaningful share of recorded sales, but revenue alone does not establish product profitability or customer loyalty.

Business implication: Product and sales teams could use the dashboard to identify high-contribution products and customers for closer review. Inventory, assortment or relationship decisions should be supported by further evidence on demand, availability, margins and repeat purchasing.

4.3 Shipping providers differed in speed and freight charges

Shipping provider	Orders	Avg. order-to-ship (days)	Avg. freight*
Federal Shipping	218	7.51	82.74
Speedy Express	206	8.54	64.95
United Package	276	9.40	81.43

*Federal Shipping had the shortest average order-to-ship time, Speedy Express had the lowest average freight, and United Package had the longest average order-to-ship time. These comparisons describe observed results; they do not prove that the provider alone caused the differences, because shipment characteristics and service conditions were not controlled for.

Business implication: Operations could review service expectations, freight charges and the types of orders handled by each provider before considering any change in allocation.

4.4 Most orders were shipped by their required dates

Approximately 95.4% of orders (668 of 700) were shipped on or before their required dates; 32 orders, or 4.6%, were shipped after their required dates. This is a measure of shipping against the required date, not proof of when customers received the orders.

Business implication: The 32 orders shipped after their required dates provide a focused starting point for further review. Actual delivery dates and reasons for delay would be needed to assess customer-facing delivery performance.

4.5 Employee contributions varied across measures

Employee order volume and revenue contributions differed. Margaret Peacock handled the highest order count in the reported analysis (146) and recorded the highest revenue contribution (approximately 242 241.27).

Fulfilment results also varied across employees. These measures should be interpreted together rather than used to rank overall employee performance in isolation, because the dataset does not account for order complexity, workload allocation or other duties.

Business implication: Managers could use these results to prompt a more contextual review of workload and fulfilment patterns, rather than treating revenue or order count as a complete measure of individual performance.

5. Recommendations

1. **Account for reporting-period coverage when reviewing sales trends:** Distinguish complete months from the final partial month when analysing order volume and revenue. Use comparable reporting periods to assess sales patterns and avoid interpreting incomplete monthly data as evidence of declining performance.
2. **Use revenue contribution to guide product reviews.** Review high-revenue products, including Côte de Blaye, alongside stock availability, demand and product margin information before making inventory or assortment decisions.
3. **Strengthen customer performance analysis.** Identify customers with high order activity and revenue contributions, then examine repeat purchasing and changes in activity over time to inform customer relationship management.
4. **Evaluate shipping providers across cost and speed.** Compare average freight charges and order-to-ship times alongside shipment characteristics before considering changes to shipping-provider allocation. The current analysis does not establish which provider offers the best overall service.
5. **Review orders shipped after their required dates.** Investigate the 32 orders shipped late to identify any recurring patterns. Where possible, incorporate actual delivery dates and recorded delay reasons to assess delivery performance more directly.
6. **Use employee metrics as a starting point for operational review.** Consider order volume, revenue contribution and fulfilment measures alongside workload and order complexity before drawing conclusions about individual performance.

6. Limitations and Future Improvements

- **Delivery outcome:** Actual delivery dates were not available, so confirmed delivery lead time and on-time delivery could not be calculated.
- **Financial context:** Product costs, operating expenses and profit margins were not available; revenue cannot be treated as profit.
- **Causation:** The analysis identifies differences and patterns but does not establish the causes behind sales or shipping outcomes.
- **Customer and employee measures:** Order and revenue contribution do not independently measure customer loyalty, retention, employee productivity or overall performance.
- **Data integration:** SQL analysis was performed in PostgreSQL while Tableau used the source Excel workbook, so the workflow was not connected through one live data source.

Future improvements should focus on adding:

- Actual delivery dates, delay reasons and more shipment-level context.
- Customer repeat-purchase and retention measures.
- Workload and order-complexity information for contextual employee analysis.
- Additional historical context for sales and product-demand investigation.
- A more integrated database-to-dashboard refresh process, where supported.

7. Conclusion

This project addressed the Northwind Traders brief by building a relational database, using SQL to answer business questions and developing an interactive Tableau dashboard. The analysis summarised sales performance, highlighted differences in product and customer contributions, compared shipping providers, assessed shipment against required dates and examined employee contributions.

The results provide management with a structured view of performance and a basis for deciding where to investigate further. The strongest next step is to pair these patterns with additional operational and financial data so that the business can move from identifying differences to understanding their causes and evaluating potential improvements.